Statistical Analysis of Atmospheric Turbulence Effects on Telescope Imaging Using Centroid Tracking

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***Abstract*—Atmospheric turbulence introduces random fluctuations in astronomical imaging, causing apparent motion of observed objects. This work analyzes centroid displacement of the Moon across video frames as a stochastic process. Statistical properties of the displacement signal are computed and compared to Gaussian models using probability density functions (PDFs), cumulative distribution functions (CDFs), and Q-Q plots. Multiple datasets with varying resolutions and durations are evaluated. The results show that the displacement is approximately Gaussian in its central region, with deviations arising from intermittency. A practical extension estimates the number of frames required for lucky imaging.**

I. Introduction

Atmospheric turbulence introduces random fluctuations in the propagation of light through the Earth's atmosphere, resulting in apparent motion and distortion in telescope images. These fluctuations arise from variations in refractive index and manifest as time-varying displacement.

In this study, the centroid position of the Moon is tracked across successive video frames and modeled as a stochastic process. The objective is to evaluate how well the displacement can be modeled using a Gaussian distribution and to analyze the effect of imaging conditions such as resolution and duration.

II. Methodology

A. Data Acquisition

Video datasets of the Moon were captured at resolutions of 240p, 480p, and 1080p with durations ranging from 2 to 5 minutes.

B. Centroid Extraction

Each frame was processed using grayscale conversion, Gaussian blur, and thresholding. The centroid was computed using image moments:

$$x_c = \frac{M_{10}}{M_{00}}, \quad y_c = \frac{M_{01}}{M_{00}} \tag{1}$$

where M_{00} is the zeroth-order image moment and M_{10} and M_{01} are the first-order moments in the horizontal and vertical directions, respectively.

C. Displacement Definition

Let $(x[n], y[n])$ denote the position of the centroid measured in frame n. The displacement of the centroid relative to a reference position (x_0, y_0) is defined as:

$$dx[n] = x[n] - x_0, \quad dy[n] = y[n] - y_0 \tag{2}$$

where (x_0, y_0) is taken as the mean centroid position over the sequence.

$$x_0 = \frac{1}{N}\sum_{n=1}^{N} x[n], \quad y_0 = \frac{1}{N}\sum_{n=1}^{N} y[n] \tag{3}$$

This removes the absolute image location and isolates the frame-to-frame centroid fluctuations caused by atmospheric turbulence and tracking variation.

D. Preprocessing

The displacement signal contains both stochastic motion and deterministic drift. To isolate the stochastic component, the following preprocessing steps were applied:

- Outlier removal using MAD threshold
- Drift removal using moving-average filtering
- Removal of corrupted segments

The slow-varying drift component was estimated using a moving-average trend signal and subtracted from the displacement signal:

$$dx_{\text{detrended}}[n] = dx[n] - \text{trend}[n] \tag{4}$$

$$dy_{\text{detrended}}[n] = dy[n] - \text{trend}_y[n] \tag{5}$$

where $\text{trend}[n]$ represents the local mean of the displacement signal over a sliding window. The moving-average trend is defined as:

$$\text{trend}[n] = \frac{1}{W}\sum_{k=-W/2}^{W/2} dx[n+k] \tag{6}$$

where W is the window length used for smoothing.

Fig. 1. Raw and detrended centroid displacement for the 480p dataset.

E. Statistical Analysis

The detrended displacement signal was treated as a stochastic process [1]. The normalized variable was defined as:

$$Z = \frac{X}{\sigma} \tag{7}$$

where X represents the detrended centroid displacement and σ is its standard deviation.

The detrended displacement signal $dx_{\text{detrended}}[n]$ is used for all subsequent statistical analysis.

III. Results

A. Time-Series Analysis

Figure 1 shows the raw and detrended centroid displacement for the 480p dataset. The raw signal exhibits a gradual drift due to imperfect telescope tracking, while the detrended signal fluctuates around zero, representing the stochastic component of atmospheric turbulence.

B. Distribution and Statistical Analysis

The distribution of the detrended centroid displacement was analyzed using histograms, empirical cumulative distribution functions (CDFs), and Q-Q plots.

As shown in Fig. 2, the 240p dataset exhibits an approximately Gaussian distribution, with good agreement between the empirical histogram and the fitted Gaussian PDF. Small deviations are observed in the tails, indicating mild non-Gaussian behavior.

The Q-Q plot in Fig. 3 reveals that the 480p dataset exhibits significant deviations from Gaussian behavior. While the central portion aligns closely with the reference line, noticeable departures occur in the tails, indicating intermittent fluctuations, which are characteristic of non-Gaussian atmospheric turbulence [2].

Figure 4 compares empirical CDFs across all datasets. The 480p dataset shows a broader distribution, while the 1080p dataset is tightly concentrated near zero. Table I summarizes the corresponding statistical properties of each dataset.

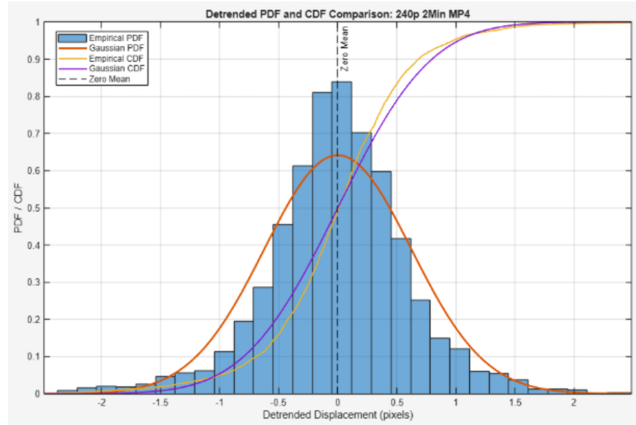

Fig. 2. PDF and CDF comparison for the 240p dataset.

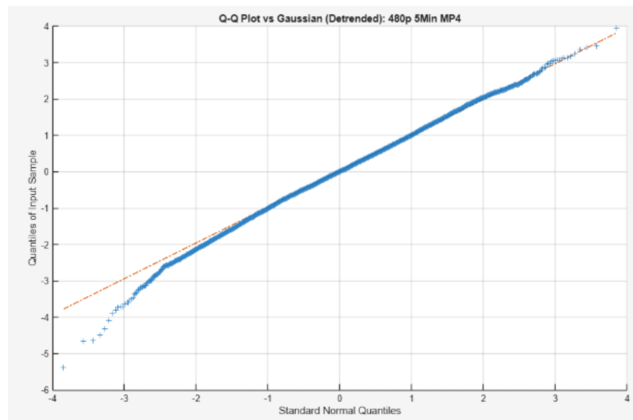

Fig. 3. Q-Q plot for the 480p dataset.

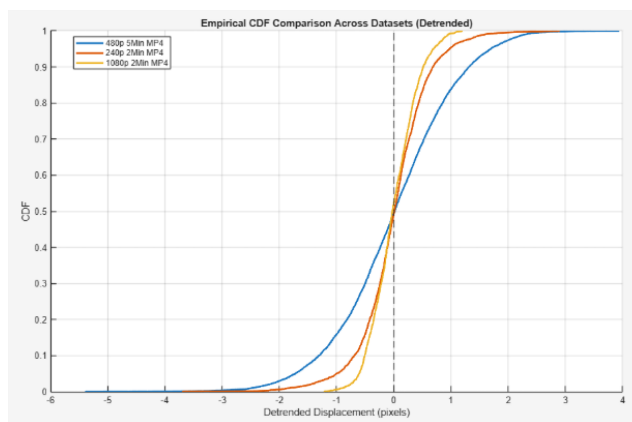

Fig. 4. CDF comparison across datasets.

TABLE I
Statistical Comparison

Dataset	σ	Skew	Kurtosis	$P(\lvert Z\rvert > 1)$
480p	1.03	-0.15	3.35	0.308
240p	0.62	-0.22	5.99	0.234
1080p	0.40	0.15	2.90	0.326

TABLE II
FRAME REQUIREMENTS

Dataset	p	Frames (avg)	Frames (95%)
480p	0.0845	3550	3878
240p	0.1664	1804	1962
1080p	0.1836	1634	1777

IV. LUCKY IMAGING ANALYSIS

Lucky imaging is a technique used in astrophotography to improve image quality by capturing a large number of short-exposure frames and selecting only the sharpest ones. Atmospheric turbulence causes random distortions that vary over time, meaning that some frames are significantly less affected than others. By discarding frames with large distortions and stacking only those with minimal motion, the final image can achieve higher resolution and clarity than would be possible with a single long exposure.

A frame is considered high-quality if the centroid displacement remains below a chosen threshold T:

$$p = P(|dx| < T) \tag{8}$$

where p represents the probability that a randomly selected frame satisfies the quality criterion. In this work, the threshold was chosen as:

$$|dx| < 0.10 \tag{9}$$

The value of p is estimated empirically from the observed displacement distribution for each dataset.

Assuming each frame is an independent realization of the stochastic process, the number of high-quality frames K obtained from N total frames follows a binomial model [1]. The expected number of usable frames is:

$$E[K] = Np \tag{10}$$

where $E[K]$ is the expected number of frames satisfying the quality condition.

To estimate the total number of frames required to obtain a target number of good frames G, the expression can be rearranged as:

$$N \approx \frac{G}{p} \tag{11}$$

This approximation provides a practical estimate of the total number of frames that must be captured to obtain a desired number of usable images for stacking. In this work, $G = 300$ usable frames was used as the target.

Using this model, the number of frames required to obtain a target number of high-quality frames can be estimated for each dataset. Table II summarizes the empirical probability of obtaining a high-quality frame and the corresponding number of total frames required to achieve 300 usable frames. The results show that higher-resolution datasets yield a greater proportion of high-quality frames.

V. DISCUSSION

The results show that centroid displacement is approximately Gaussian in its central region but exhibits deviations in the tails due to turbulence intermittency. This behavior is consistent with prior studies showing that atmospheric turbulence can exhibit non-Gaussian characteristics, particularly in extreme events [2]. Higher resolution reduces variance and improves agreement with the Gaussian model, while lower resolution introduces increased intermittency.

VI. CONCLUSION

This work demonstrates that atmospheric turbulence can be modeled as a stochastic process with approximately Gaussian behavior in its central region, with deviations arising in the tails due to intermittency. A practical extension shows that thousands of frames are required for effective lucky imaging, with higher-resolution data reducing the number of frames needed. These findings provide a statistical foundation for the effectiveness of lucky imaging techniques.

REFERENCES

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[2] P. M. Reeves, R. G. Joppa, and V. M. Ganzer, “A non-Gaussian model of continuous atmospheric turbulence for use in aircraft design,” NASA Contractor Report NASA-CR-2639, 1976.